1. Write a Java program to find the area of a triangle given its three sides using Heron's formula.

```
import java.util.Scanner;

public class TriangleArea {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input the three sides
        System.out.print("Enter side 1: ");
        double a = sc.nextDouble();

        System.out.print("Enter side 2: ");
        double b = sc.nextDouble();

        System.out.print("Enter side 3: ");
        double c = sc.nextDouble();

        // Calculate semi-perimeter
        double s = (a + b + c) / 2;

        // Heron's formula
        double area = Math.sqrt(s * (s - a) * (s - b) * (s -
c));

        // Display result
        System.out.println("Area of the triangle = " + area);

        sc.close();
    }
}
```

Formula;

s=a+b+c2

Area=s(s−a)(s−b)(s−c)

Output

For input sides 3, 4, and 5:

```
Enter side 1: 3
Enter side 2: 4
```

```
Enter side 3: 5
Area of the triangle = 6.0
```

2. Create a Java program to calculate the compound interest given principal, rate, and time.

```
import java.util.Scanner;

public class CompoundInterest {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Input principal, rate and time
        System.out.print("Enter principal amount: ");
        double p = sc.nextDouble();

        System.out.print("Enter rate of interest: ");
        double r = sc.nextDouble();

        System.out.print("Enter time in years: ");
       double t = sc.nextDouble();

        // Calculate amount and compound interest
        double amount = p * Math.pow((1 + r / 100), t);
        double compoundInterest = amount - p;

        System.out.println("Compound Interest = " +
compoundInterest);
        System.out.println("Total Amount = " + amount);

        sc.close();
    }
}
```

The formula used is:

A=P(1+R100)T

CI=A−P

Example Output

```
Enter principal amount: 10000
Enter rate of interest: 10
Enter time in years: 2
Compound Interest = 2100.0
```

```
Total Amount = 12100.0
```

3. Develop a Java program to find the distance between two points in a 2D plane using the distance formula.

```java
import java.util.Scanner;

public class DistanceBetweenPoints {
   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      // Input coordinates of first point
      System.out.print("Enter x1: ");
      double x1 = sc.nextDouble();

      System.out.print("Enter y1: ");
      double y1 = sc.nextDouble();

      // Input coordinates of second point
      System.out.print("Enter x2: ");
      double x2 = sc.nextDouble();

      System.out.print("Enter y2: ");
      double y2 = sc.nextDouble();

      // Calculate distance
      double distance = Math.sqrt(
         Math.pow(x2 - x1, 2) + Math.pow(y2 - y1, 2)
      );

      System.out.println("Distance between the two points = " + distance);

      sc.close();
   }
}
```

Output

```
Enter x1: 2
Enter y1: 3
Enter x2: 5
Enter y2: 7
Distance between the two points = 5.0
```

4. Write a Java program to demonstrate prefix and postfix increment by simulating a counter for visitors entering and leaving a store.

```
public class VisitorCounter {
    public static void main(String[] args) {
        int visitors = 0;

        // Visitor enters the store - postfix increment
        System.out.println("Visitor enters: " + visitors++);
        System.out.println("Visitors inside: " + visitors);

        // Another visitor enters - prefix increment
        System.out.println("Visitor enters: " + ++visitors);
        System.out.println("Visitors inside: " + visitors);

        // Visitor leaves - postfix decrement
        System.out.println("Visitor leaves: " + visitors--);
        System.out.println("Visitors inside: " + visitors);

        // Another visitor leaves - prefix decrement
        System.out.println("Visitor leaves: " + --visitors);
        System.out.println("Visitors inside: " + visitors);
    }
}
```

Output

```
Visitor enters: 0
Visitors inside: 1
Visitor enters: 2
Visitors inside: 2
Visitor leaves: 2
Visitors inside: 1
Visitor leaves: 0
Visitors inside: 0
```

5. Create a Java program that takes an integer and outputs its negative value using unary operators.

```
import java.util.Scanner;

public class NegativeValue {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int num = sc.nextInt();
```

```java
        // Using unary minus operator
        int negative = -num;

        System.out.println("Negative value = " + negative);

        sc.close();
    }
}
```

Output

```
Enter an integer: 25
Negative value = -25
```

6. Write a Java program to gradually reduce a number by half using /= until it becomes less than 1, counting the number of steps.

```java
import java.util.Scanner;

public class ReduceByHalf {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        double num = sc.nextDouble();

        int steps = 0;

        while (num >= 1) {
            num /= 2;
            steps++;
        }

        System.out.println("Number of steps: " + steps);
        System.out.println("Final value: " + num);
    }
}
```

Output:
Enter a number: 16
Number of steps: 5
Final value: 0.5

7. Create a Java program to accumulate rainfall data for 7 days using += and find the total.

```
public class Rainfall {
    public static void main(String[] args) {
        double total = 0;

        for (int day = 1; day <= 7; day++) {
            double rainfall = day * 2; // sample rainfall
            total += rainfall;
        }

        System.out.println("Total rainfall for 7 days: " + total + " mm");
    }
}
```

Output:

```
Total rainfall for 7 days: 56.0 mm
```

8. Write a Java program to check whether three given angles form a valid triangle.

```
import java.util.Scanner;

public class TriangleAngles {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first angle: ");
        double angle1 = sc.nextDouble();

        System.out.print("Enter second angle: ");
        double angle2 = sc.nextDouble();

        System.out.print("Enter third angle: ");
        double angle3 = sc.nextDouble();

        if (angle1 > 0 && angle2 > 0 && angle3 > 0 &&
            angle1 + angle2 + angle3 == 180) {
            System.out.println("The angles form a valid triangle.");
        } else {
            System.out.println("The angles do not form a valid triangle.");
        }

        sc.close();
    }
}
```

Output-
Enter first angle: 60
Enter second angle: 60
Enter third angle: 60
The angles form a valid triangle.

9. Create a Java program to compare two strings for lexicographic order without using built-in compare functions (use relational operators on characters).

```
import java.util.Scanner;

public class LexicographicCompare {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first string: ");
        String str1 = sc.nextLine();

        System.out.print("Enter second string: ");
        String str2 = sc.nextLine();

        int minLength = Math.min(str1.length(), str2.length());
        int result = 0;

        for (int i = 0; i < minLength; i++) {
            if (str1.charAt(i) < str2.charAt(i)) {
                result = -1;
                break;
            } else if (str1.charAt(i) > str2.charAt(i)) {
                result = 1;
                break;
            }
        }

        if (result == 0) {
            if (str1.length() < str2.length()) {
                result = -1;
            } else if (str1.length() > str2.length()) {
                result = 1;
            }
        }

        if (result < 0) {
            System.out.println(str1 + " comes before " + str2);
        } else if (result > 0) {
            System.out.println(str1 + " comes after " + str2);
        } else {
            System.out.println("Both strings are equal.");
```

```java
        }

        sc.close();
    }
}
```

Output-
Enter first string: Apple
Enter second string: Banana
Apple comes before Banana

10. Write a Java program to determine if a given year is both a leap year and within a given range.

```java
import java.util.Scanner;

public class LeapYearRange {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter year: ");
        int year = sc.nextInt();

        System.out.print("Enter lower range: ");
        int lower = sc.nextInt();

        System.out.print("Enter upper range: ");
        int upper = sc.nextInt();

        if ((year % 400 == 0 || (year % 4 == 0 && year % 100 != 0))
                && year >= lower && year <= upper) {
            System.out.println(year + " is a leap year and is within the given range.");
        } else {
            System.out.println(year + " does not satisfy both conditions.");
        }

        sc.close();
    }
}
```

Output-
Enter year: 2024
Enter lower range: 2000
Enter upper range: 2030
2024 is a leap year and is within the given range.

11. Create a Java program to check if a student passes a course, where they must have at least 40% in theory AND 50% in practical OR overall, 50%.

```java
import java.util.Scanner;

public class StudentPass {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter theory marks: ");
        double theory = sc.nextDouble();

        System.out.print("Enter practical marks: ");
        double practical = sc.nextDouble();

        System.out.print("Enter total marks: ");
        double total = sc.nextDouble();

        double overall = (theory + practical) / total * 100;

        if ((theory >= 40 && practical >= 50) || overall >= 50) {
            System.out.println("Student passes the course.");
        } else {
            System.out.println("Student fails the course.");
        }

        sc.close();
    }
}
```

Output-

```
Enter theory marks: 45
Enter practical marks: 55
Enter total marks: 200
Student passes the course.
```

12. Write a Java program to find the smallest of four numbers using nested ternary operators.

```java
import java.util.Scanner;

public class SmallestOfFour {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter four numbers: ");
        int a = sc.nextInt();
        int b = sc.nextInt();
        int c = sc.nextInt();
        int d = sc.nextInt();
```

```
        int smallest = (a < b)
                ? ((a < c) ? ((a < d) ? a : d) : ((c < d) ? c : d))
                : ((b < c) ? ((b < d) ? b : d) : ((c < d) ? c : d));

        System.out.println("Smallest number = " + smallest);

        sc.close();
    }
}
```

Output-
Enter four numbers: 25 12 18 30
Smallest number = 12

13. Create a Java program to check if a character is a vowel, consonant, digit, or special symbol using the ternary operator.

```
import java.util.Scanner;

public class CharacterCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a character: ");
        char ch = sc.next().charAt(0);

        String result = (ch == 'a' || ch == 'e' || ch == 'i' ||
                         ch == 'o' || ch == 'u' ||
                         ch == 'A' || ch == 'E' || ch == 'I' ||
                         ch == 'O' || ch == 'U')
                        ? "Vowel"
                        : ((ch >= '0' && ch <= '9')
                           ? "Digit"
                           : ((ch >= 'A' && ch <= 'Z') ||
                              (ch >= 'a' && ch <= 'z'))
                             ? "Consonant"
                             : "Special Symbol");

        System.out.println("The character is a " + result + ".");

        sc.close();
    }
}
```

Output

```
Enter a character: E
The character is a Vowel.
```

Another example:

```
Enter a character: 7
The character is a Digit.

Enter a character: #
The character is a Special Symbol.
```

14. Write a Java program to swap two integers without using a temporary variable (use XOR).

```
import java.util.Scanner;

public class SwapUsingXOR {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter first integer: ");
        int a = sc.nextInt();

        System.out.print("Enter second integer: ");
        int b = sc.nextInt();

        // Swapping using XOR
        a = a ^ b;
        b = a ^ b;
        a = a ^ b;

        System.out.println("After swapping:");
        System.out.println("First integer = " + a);
        System.out.println("Second integer = " + b);

        sc.close();
    }
}
```

Output-

Enter first integer: 10
Enter second integer: 20
After swapping:
First integer = 20
Second integer = 10

15. Create a Java program to count the number of set bits (1s) in the binary representation of a number using bitwise operators.

```
import java.util.Scanner;
```

```java
public class CountSetBits {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        int count = 0;
        int n = num;

        while (n != 0) {
            count += n & 1;  // Check the last bit
            n = n >> 1;      // Right shift by 1 bit
        }

        System.out.println("Number of set bits = " + count);

        sc.close();
    }
}
```

Output-

```
Enter a number: 15
Number of set bits = 4
```

16. Write a Java program to perform fast multiplication and division of a number by powers of two using shift operators.

```java
import java.util.Scanner;

public class ShiftOperations {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int num = sc.nextInt();

        System.out.print("Enter the power of 2: ");
        int power = sc.nextInt();

        // Multiplication by 2^power
        int multiplication = num << power;

        // Division by 2^power
        int division = num >> power;

        System.out.println("After multiplication by 2^" + power + " = " + multiplication);
```

```java
        System.out.println("After division by 2^" + power + " = " + division);

        sc.close();
    }
}
```

Output-
Enter a number: 8
Enter the power of 2: 2
After multiplication by 2^2 = 32
After division by 2^2 = 2

17. Create a Java program to cyclically rotate the bits of an integer to the left by 2 positions.

```java
import java.util.Scanner;

public class LeftRotate {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int num = sc.nextInt();

        // Cyclic left rotation by 2 positions
        int rotated = (num << 2) | (num >>> 30);

        System.out.println("After left rotation by 2 positions = " + rotated);

        sc.close();
    }
}
```

Output-
Enter an integer: 10
After left rotation by 2 positions = 40

18. Write a Java program to print the first 20 terms of the Fibonacci sequence using a for loop.

```java
public class FibonacciSeries {
    public static void main(String[] args) {
        int first = 0;
        int second = 1;

        System.out.println("First 20 terms of Fibonacci sequence:");
```

```
        for (int i = 1; i <= 20; i++) {
            System.out.print(first + " ");

            int next = first + second;
            first = second;
            second = next;
        }
    }
}
```

Output-
First 20 terms of Fibonacci sequence:
0 1 1 2 3 5 8 13 21 34 55 89 144 233 377 610 987 1597 2584 4181

19. Create a Java program to print all Armstrong numbers between 1 and 1000 using a for loop.

```
public class ArmstrongNumbers {
    public static void main(String[] args) {

        for (int num = 1; num <= 1000; num++) {
            int temp = num;
            int sum = 0;

            while (temp > 0) {
                int digit = temp % 10;
                sum += digit * digit * digit;
                temp /= 10;
            }

            if (sum == num) {
                System.out.println(num);
            }
        }
    }
}
```

Output-
1
153
370
371
407

20. Write a Java program to repeatedly accept a password until the correct one is entered (use do-while).

```java
import java.util.Scanner;

public class PasswordCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String password;
        String correctPassword = "java123";

        do {
            System.out.print("Enter password: ");
            password = sc.nextLine();

            if (!password.equals(correctPassword)) {
                System.out.println("Incorrect password. Try again.");
            }

        } while (!password.equals(correctPassword));

        System.out.println("Correct password! Access granted.");

        sc.close();
    }
}
```

Output-

```
Enter password: hello
Incorrect password. Try again.
Enter password: 12345
Incorrect password. Try again.
Enter password: java123
Correct password! Access granted.
```

21. Create a Java program to display all factors of a number using a do-while loop.

```java
import java.util.Scanner;

public class Factors {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int i = 1;

        System.out.println("Factors of " + n + " are:");
```

```
        do {
            if (n % i == 0) {
                System.out.println(i);
            }
            i++;
        } while (i <= n);

        sc.close();
    }
}
```

Output-
Factors of 12 are:
1
2
3
4
6
12

22. Write a Java program to reverse the digits of an integer using a while loop.

```
import java.util.Scanner;

public class ReverseNumber {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter an integer: ");
        int n = sc.nextInt();

        int reverse = 0;

        while (n != 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        System.out.println("Reversed number = " + reverse);

        sc.close();
    }
}
```

input: 12345

Output: 54321

23. Create a Java program to determine if a number is a palindrome using a while loop.

```
import java.util.Scanner;

public class Palindrome {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        int original = n;
        int reverse = 0;

        while (n != 0) {
            int digit = n % 10;
            reverse = reverse * 10 + digit;
            n = n / 10;
        }

        if (original == reverse) {
            System.out.println("The number is a palindrome.");
        } else {
            System.out.println("The number is not a palindrome.");
        }

        sc.close();
    }
}
```

Input: 121
Output: The number is a palindrome.

24. Write a Java program to find the maximum and minimum values in an integer array using a for-each loop.

```
import java.util.Scanner;

public class MaxMinArray {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
```

```
        int[] arr = new int[n];

        System.out.println("Enter the elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }

        int max = arr[0];
        int min = arr[0];

        for (int num : arr) {
            if (num > max) {
                max = num;
            }

            if (num < min) {
                min = num;
            }
        }

        System.out.println("Maximum value = " + max);
        System.out.println("Minimum value = " + min);

        sc.close();
    }
}
```

Input: 10 25 5 40 15

Output:

```
Maximum value = 40
Minimum value = 5
```

25. Create a Java program to calculate the average marks of students stored in a 2D array using a for-each loop.

```
import java.util.Scanner;

public class StudentAverage {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of students: ");
        int students = sc.nextInt();

        System.out.print("Enter number of subjects: ");
        int subjects = sc.nextInt();
```

```
        int[][] marks = new int[students][subjects];

        System.out.println("Enter marks:");
        for (int i = 0; i < students; i++) {
            System.out.println("Student " + (i + 1) + ":");
            for (int j = 0; j < subjects; j++) {
                marks[i][j] = sc.nextInt();
            }
        }

        int sum = 0;
        int count = 0;

        for (int[] student : marks) {
            for (int mark : student) {
                sum += mark;
                count++;
            }
        }

        double average = (double) sum / count;

        System.out.println("Average marks = " + average);

        sc.close();
    }
}
```

For 2 students with marks:

```
Student 1: 80 70 90
Student 2: 60 75 85
```

Output:

```
Average marks = 76.66666666666667
```

26. Write a Java program that takes a day number (1–7) and displays whether it's a weekday or weekend using switch-case.

```
import java.util.Scanner;

public class DayType {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter day number (1-7): ");
        int day = sc.nextInt();
```

```
        switch (day) {
            case 1:
            case 2:
            case 3:
            case 4:
            case 5:
                System.out.println("Weekday");
                break;

            case 6:
            case 7:
                System.out.println("Weekend");
                break;

            default:
                System.out.println("Invalid day number");
        }

        sc.close();
    }
}
```

Input: 6
Output:

```
Weekend
```

27. Create a Java program that takes a grade (A–F) and prints its qualitative meaning (e.g., A → Excellent, B → Good, etc.) using switch-case.

```
import java.util.Scanner;

public class GradeMeaning {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter grade (A-F): ");
        char grade = sc.next().toUpperCase().charAt(0);

        switch (grade) {
            case 'A':
                System.out.println("Excellent");
                break;
            case 'B':
                System.out.println("Good");
                break;
```

```
            case 'C':
                System.out.println("Average");
                break;
            case 'D':
                System.out.println("Below Average");
                break;
            case 'E':
                System.out.println("Pass");
                break;
            case 'F':
                System.out.println("Fail");
                break;
            default:
                System.out.println("Invalid grade");
        }

        sc.close();
    }
}
```

Output

```
Enter grade (A-F): A
Excellent
```

Another example:

```
Enter grade (A-F): C
Average
```

For F:

```
Enter grade (A-F): F
Fail
```

28. Write a Java program to print numbers from 1 to 50, skipping all numbers that are perfect squares using continue.

```
public class PerfectSquareSkip {
    public static void main(String[] args) {
        for (int i = 1; i <= 50; i++) {
            int root = (int) Math.sqrt(i);

            if (root * root == i) {
                continue;
            }
```

```
            System.out.print(i + " ");
        }
    }
}
```

Output-
2 3 5 6 7 8 10 11 12 13 14 15 17 18 19 20 21 22 23 24 26 27 28 29 30 31 32 33 34 35 37 38 39 40 41 42 43 44 45 46 47 48 50

29. Create a Java program to generate random integers between 1 and 100 and stop when a number divisible by both 7 and 13 is found using break.

```
import java.util.Random;

public class RandomNumber {
    public static void main(String[] args) {
        Random rand = new Random();

        while (true) {
            int num = rand.nextInt(100) + 1;
            System.out.println("Generated: " + num);

            if (num % 7 == 0 && num % 13 == 0) {
                System.out.println("Number divisible by both 7 and 13: " + num);
                break;
            }
        }
    }
}
```

Output-
Generated: 23
Generated: 56
Generated: 41
Generated: 78
Generated: 91
Number divisible by both 7 and 13: 91

30. Write a Java program that: Uses shift operators to check if a number is a power of 4, Uses bitwise to toggle the 3rd bit of that number, Uses a for loop to print its multiplication table but skips multiples of 6 using continue, Stops if it reaches a multiple of 48 using break.

```
import java.util.Scanner;

public class PowerOfFour {
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.print("Enter a number: ");
    int num = sc.nextInt();

    // Check if number is a power of 4 using shift operators
    int temp = num;
    boolean isPowerOf4 = false;

    while (temp > 1 && (temp & 3) == 0) {
        temp >>= 2;
    }

    if (temp == 1) {
        isPowerOf4 = true;
    }

    if (isPowerOf4) {
        System.out.println(num + " is a power of 4.");
    } else {
        System.out.println(num + " is not a power of 4.");
    }

    // Toggle the 3rd bit using XOR
    // 3rd bit means bit position 2
    int toggled = num ^ (1 << 2);

    System.out.println("After toggling 3rd bit: " + toggled);

    // Multiplication table
    System.out.println("Multiplication table:");

    for (int i = 1; i <= 20; i++) {
        int result = num * i;

        if (result % 6 == 0) {
            continue;
        }

        if (result % 48 == 0) {
            break;
        }

        System.out.println(num + " x " + i + " = " + result);
    }

    sc.close();
```

```
    }
}
```

Output-

Enter a number: 4
4 is a power of 4.
After toggling 3rd bit: 0
Multiplication table:
4 x 1 = 4
4 x 2 = 8
4 x 4 = 16
4 x 5 = 20
4 x 7 = 28
4 x 8 = 32
4 x 10 = 40
4 x 11 = 44